

CPSC 304 Project Cover Page

Milestone #: 4

Date: August 5, 2024

Group Number: 25

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia.

2. SQL script available in repo as db.sql

3.

a) This ski resort management system was developed to aid in the administrative functions of a ski resort. The application allows for the management of various entities of a ski resort like guest, classes, and runs/lifts, providing functionalities to search, update, and delete entries efficiently. It further supports more complex SQL operations such as joins, divisions, and aggregations. Users can interact with the system through a user-friendly graphical interface. The program has a variety of queries across different entities that reflect the operational needs of our resort integrated through a backend system with Oracle Database. The final deliverable is a functional GUI that allows easy access to the database, ensuring that all user actions result in immediate and visible changes within the system.

b) Fixed the primary keys of Take to include PKs of both entities.

c) Patron(patronID, name, email)

SkiInstructor(employeeID, name, availability)

SkiLift(liftNumber, capacity, status)

Class(Date, classLevel, capacity, **InstructorID**, **liftNumber**)

Take(**patronID**, **Date**, **classLevel**)

Rental2(**patronID**, height)

Rental3(Date, patronID, equipmentType)

Rental4(**patronID**, footSize)

SkiRun(runNumber, difficulty, status)

LiftTicket2(discountType, discountPercentage)

LiftTicket1(ticketNo, **discountType**, **patronID**)

DayTicket(**ticketNo**, Date)

SeasonPass(**ticketNo**, season)

Restaurant(name, capacity, timing)

EatsAt(**patronID**, **restaurantName**)

Discount(**ticketNo**, **restaurantName**)

Accesses(**runNumber**, **liftNumber**)

Patron:

```
SQL> select * from Patron;
```

PATRONID	NAME	EMAIL
P001	Jake John	jjohn2@gmail.com
P002	Lili Henrik	lilihenn@gmail.com
P003	Jasper Deasey	jasperstezy@gmail.com
P004	Ella Krav	ellakrav@gmail.com
P005	Blake Smith	smithblake@gmail.com
P006	Emily Johnson	emily.j@gmail.com
P007	Sarah Brown	sbrown@gmail.com
P008	David Wilson	dave.r.wilson@gmail.com
P009	Jessica Martinez	jess.mrtnz@gmail.com

SkiInstructor:

EMPLOYEEID	NAME	AVAILABILITY
E001	Instructor One	Available
E002	Instructor Two	Available
E003	Instructor Three	Available
E004	Instructor Four	Available
E005	Instructor Five	Unavailable

SkiLift:

LIFTNUMBER	CAPACITY	STATUS
L001	4	Operational
L002	6	Operational
L003	4	Under Maintenance
L004	4	Operational
L005	6	Operational
L006	4	Under Maintenance

Class:

Date	CLASSLEVEL	CAPACITY	INSTRUCTORID	LIFTNUMBER
2024-12-20	Beginner	5	E001	L001
2024-12-20	Intermediate	10	E001	L002
2024-12-21	Advanced	10	E003	L004
2024-12-22	Beginner	10	E001	L001
2024-12-22	Intermediate	10	E004	L002
2024-12-23	Intermediate	3	E005	L005
2024-12-27	Advanced	5	E003	L002
2024-12-27	Beginner	10	E002	L004

Take:

PATRONID	Date	CLASSLEVEL
P001	2024-12-20	Beginner
P001	2024-12-22	Beginner
P001	2024-12-27	Beginner
P002	2024-12-21	Advanced
P002	2024-12-27	Advanced
P003	2024-12-20	Beginner
P003	2024-12-22	Intermediate
P003	2024-12-27	Beginner
P004	2024-12-20	Intermediate
P004	2024-12-22	Intermediate
P004	2024-12-27	Beginner
P005	2024-12-22	Intermediate
P005	2024-12-27	Beginner
P006	2024-12-20	Beginner
P006	2024-12-22	Intermediate
P006	2024-12-27	Beginner
P007	2024-12-20	Beginner
P007	2024-12-22	Intermediate
P007	2024-12-23	Intermediate
P007	2024-12-27	Beginner
P008	2024-12-27	Advanced
P009	2024-12-23	Intermediate
P009	2024-12-27	Beginner

Rental2:

PATRONID	HEIGHT
P001	5.8
P004	6.1
P005	5.8

Rental3:

Date	PATRONID	EQUIPMENTTYPE
2024-12-20	P001	Skis+Boots
2024-12-22	P001	Skis+Boots
2024-12-20	P004	Skis+Boots
2024-12-22	P005	Boots
2024-12-27	P005	Boots

Rental4:

PATRONID	FOOTSIZE
P001	9.5
P004	12
P005	8

SkiRun:

RUNNUMBER	DIFFICULTY	STATUS
R001	Green	Open
R002	Green	Open
R003	Black	Closed
R004	Double Black	Open
R005	Blue	Open
R006	Black	Open

LiftTicket2:

DISCOUNTTYPE	DISCOUNTPERCENTAGE
None	0
Student	25
Senior	30

LiftTicket1:

TICKETNO	DISCOUNTTYPE	PATRONID
D001	None	P001
D002	None	P002
D003	Student	P003
D004	None	P004
D005	Senior	P005
S001	Student	P006
S002	None	P007
S003	Senior	P008
S004	None	P009
S005	None	P004

DayTicket:

TICKETNO	Date
D001	2024-12-20
D002	2024-12-21
D003	2024-12-22
D004	2024-12-20
D005	2024-12-21

SeasonPass:

TICKETNO	SEASON
S001	2024-2025
S002	2024-2025
S003	2024-2025
S004	2023-2024
S005	2023-2024

Restaurant:

NAME	CAPACITY	TIMING
Mountain Top	100	09:00-21:00
Creekside	75	11:00-20:00
Glacier Hut	50	09:00-15:00
Peak View	50	09:00-15:00
Round House	250	07:00-17:00

EatsAt:

PATRONID	RESTAURANTNAME
P002	Round House
P003	Creekside
P003	Glacier Hut
P004	Mountain Top
P005	Creekside

Discount:

TICKETNO	RESTAURANTNAME
S002	Creekside
S002	Glacier Hut
S002	Mountain Top
S003	Creekside
S003	Glacier Hut
S003	Mountain Top

Accesses:

RUNNUMBER	LIFTNUMBER
R001	L001
R001	L006
R002	L001
R002	L006
R003	L003
R003	L006
R004	L004
R005	L005
R006	L002

d) All SQL queries can be found in source/appService.js

Insert: Line 349

Delete: Line 189-201

Update: Line 382

Selection: Line 363-376

Projection: Line 147-159

Join: Line 282-288

Aggregation with GROUP BY: Line 227

Aggregation with HAVING: Line 240

Nested Aggregation with GROUP BY: Line 214

Division: Line: 299-313

e)

Insert: Line 349

Before (Browse Record Page):

RENTAL3

Date

☒

mm/dd/yyyy

PATRONID

☒

EQUIPMENTTYPE

☒

Find

Date	PATRONID	EQUIPMENTTYPE	Actions
2024-12-20T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-20T07:00:00.000Z	P004	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P005	Boots	Delete
2024-12-27T07:00:00.000Z	P005	Boots	Delete

During (Add Rental Page)

Rentals

Patron

P001

Date

12/26/2024

Equipment

Helmet

Create Rental

After (Browse Record Page):

RENTAL3

▼

Date

☒

mm/dd/yyyy

PATRONID

☒

EQUIPMENTTYPE

☒

Find

Date	PATRONID	EQUIPMENTTYPE	Actions
2024-12-26T07:00:00.000Z	P001	Helmet	Delete
2024-12-20T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-20T07:00:00.000Z	P004	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P005	Boots	Delete
2024-12-27T07:00:00.000Z	P005	Boots	Delete

Delete: Line 189-201

Before (Browse Records)

RENTAL3

▼

Date

☒

mm/dd/yyyy

PATRONID

☒

EQUIPMENTTYPE

☒

Find

Date	PATRONID	EQUIPMENTTYPE	Actions
2024-12-26T07:00:00.000Z	P001	Helmet	Delete
2024-12-20T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-20T07:00:00.000Z	P004	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P005	Boots	Delete
2024-12-27T07:00:00.000Z	P005	Boots	Delete

During:

localhost:65535 says
Deleted.
OK

RENTAL3

☒ mm/dd/yyyy

PATRONID

☒

EQUIPMENTTYPE

☒

Find

Date	PATRONID	EQUIPMENTTYPE	Actions
2024-12-26T07:00:00.000Z	P001	Helmet	Delete
2024-12-20T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-20T07:00:00.000Z	P004	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P005	Boots	Delete
2024-12-27T07:00:00.000Z	P005	Boots	Delete

After:

RENTAL3

▼

Date

☒

mm/dd/yyyy



PATRONID

☒

EQUIPMENTTYPE

☒

Find

Date	PATRONID	EQUIPMENTTYPE	Actions
2024-12-20T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P001	Skis+Boots	Delete
2024-12-20T07:00:00.000Z	P004	Skis+Boots	Delete
2024-12-22T07:00:00.000Z	P005	Boots	Delete
2024-12-27T07:00:00.000Z	P005	Boots	Delete

Update: Line 382

Before (Patron Page)

Custom Search and Update

Criteria

Search

Patron ID	Name	Email	Action
P001	Jake John	jjohn2@gmail.com	Update
P002	Lili Henrik	lilihenn@gmail.com	Update
P003	Jasper Deasey	jasperstezy@gmail.com	Update
P004	Ella Krav	ellakrav@gmail.com	Update
P005	Blake Smith	smithblake@gmail.com	Update
P006	Emily Johnson	emily.j@gmail.com	Update
P007	Sarah Brown	sbrown@gmail.com	Update
P008	David Wilson	dave.r.wilson@gmail.com	Update

During:

Custom Search and Update

Criteria

Search

Patron ID	Name	Email	Action
P001	Jake John	jjohn2@gmail.com	<button>Update</button>
P002	Lili Henrik	lilihenn@gmail.com	<button>Update</button>
P003	Jasper Deasey	jasperstezy@gmail.com	<button>Update</button>
P004	Ella Krav	ellakrav@gmail.com	<button>Update</button>
P005	Blake Smith	smithblake@gmail.com	<button>Update</button>
P006	Emily Johnson	emily.j@gmail.com	<button>Update</button>
P007	Sarah Brown	sbrown@gmail.com	<button>Update</button>
P008	David Wilson	dave.r.wilson@gmail.com	<button>Update</button>

Jonny John John

jjohn2@gmail.com

Save Changes

Cancel

After:

Custom Search and Update

Criteria

Search

Patron ID	Name	Email	Action
P001	Jonny John John	jjohn2@gmail.com	Update
P002	Lili Henrik	lilihenn@gmail.com	Update
P003	Jasper Deasey	jasperstezy@gmail.com	Update
P004	Ella Krav	ellakrav@gmail.com	Update
P005	Blake Smith	smithblake@gmail.com	Update
P006	Emily Johnson	emily.j@gmail.com	Update
P007	Sarah Brown	sbrown@gmail.com	Update
P008	David Wilson	dave.r.wilson@gmail.com	Update

Selection: Line 363-376

Before (Patron Page):

Custom Search and Update

To use the custom search, use the syntax "Patron ID == "PXXX" || Name == "Alan" && email >= "alan@gmail.com"..."

Criteria

Search			
Patron ID	Name	Email	Action
P001	Jonny John John	jjohn2@gmail.com	Update
P002	Lili Henrik	lilihenn@gmail.com	Update
P003	Jasper Deasey	jasperstezy@gmail.com	Update
P004	Ella Krav	ellakrav@gmail.com	Update
P005	Blake Smith	smithblake@gmail.com	Update
P006	Emily Johnson	emily.j@gmail.com	Update
P007	Sarah Brown	sbrown@gmail.com	Update
P008	David Wilson	dave.r.wilson@gmail.com	Update

During:

Custom Search and Update

To use the custom search, use the syntax "Patron ID == "PXXX" || Name == "Alan" && email == "alan@gmail.com"..."

Criteria

patronID == "P001" && Name == "Jonny John John"

Search			
Patron ID	Name	Email	Action
P001	Jonny John John	jjohn2@gmail.com	<button>Update</button>
P002	Lili Henrik	lilihenn@gmail.com	<button>Update</button>
P003	Jasper Deasey	jasperstezy@gmail.com	<button>Update</button>
P004	Ella Krav	ellakrav@gmail.com	<button>Update</button>
P005	Blake Smith	smithblake@gmail.com	<button>Update</button>
P006	Emily Johnson	emily.j@gmail.com	<button>Update</button>
P007	Sarah Brown	sbrown@gmail.com	<button>Update</button>
P008	David Wilson	dave.r.wilson@gmail.com	<button>Update</button>

After:

Custom Search and Update

To use the custom search, use the syntax "Patron ID == "PXXX" || Name == "Alan" && email >= "alan@gmail.com"..."

Criteria

patronID == "P001" && Name == "Jonny John John"

Search

Patron ID	Name	Email	Action
P001	Jonny John John	jjohn2@gmail.com	Update

Projection: Line 147-159

Before (Browse Record Page):

PATRON ☐

PATRONID



NAME



EMAIL



Find

PATRONID	NAME	EMAIL	Actions
P001	Jonny John John	jjohn2@gmail.com	Delete
P002	Lili Henrik	lilihenn@gmail.com	Delete
P003	Jasper Deasey	jasperstezy@gmail.com	Delete
P004	Ella Krav	ellakrav@gmail.com	Delete
P005	Blake Smith	smithblake@gmail.com	Delete
P006	Emily Johnson	emily.j@gmail.com	Delete
P007	Sarah Brown	sbrown@gmail.com	Delete

During:

PATRON

▼

PATRONID

☒

NAME

☐

EMAIL

☐

Find

PATRONID	NAME	EMAIL	Actions
P001	Jonny John John	jjohn2@gmail.com	Delete
P002	Lili Henrik	lilihenn@gmail.com	Delete
P003	Jasper Deasey	jasperstezy@gmail.com	Delete
P004	Ella Krav	ellakrav@gmail.com	Delete
P005	Blake Smith	smithblake@gmail.com	Delete
P006	Emily Johnson	emily.j@gmail.com	Delete
P007	Sarah Brown	sbrown@gmail.com	Delete

After:

PATRON

PATRONID

☒

NAME

☐

EMAIL

☐

Find

PATRONID	Actions
P001	Delete
P002	Delete
P003	Delete
P004	Delete
P005	Delete
P006	Delete
P007	Delete

Join: Line 282-288

Before (Runs and Lifts):

[Find Runs for Lift](#)

Black

▼

[Find Lifts](#)

Lifts

Show All Lifts		
Lift Number	Chair Capacity	Status
L001	4	Operational
L002	6	Operational
L003	4	Under Maintenance
L004	4	Operational
L005	6	Operational
L006	4	Under Maintenance

Runs

Run Difficulty:

All

▼

Run Status:

All

▼

[Search](#)

Show All Runs		
Run Number	Difficulty	Status
R001	Green	Open
R002	Green	Open
R003	Black	Closed

During:

Find Runs for Lift

Black

Find Lifts

Lifts

Show All Lifts		
Lift Number	Chair Capacity	Status
L001	4	Operational
L002	6	Operational
L003	4	Under Maintenance
L004	4	Operational
L005	6	Operational
L006	4	Under Maintenance

After:

Find Runs for Lift

Lift Number	Chair Capacity	Status	Run Number	Difficulty	Status
L001	4	Operational	R001	Green	Open
L001	4	Operational	R002	Green	Open

Black

▼

Find Lifts

Lifts

Show All Lifts

Lift Number	Chair Capacity	Status
L001	4	Operational
L002	6	Operational
L003	4	Under Maintenance
L004	4	Operational
L005	6	Operational
L006	4	Under Maintenance

Aggregation with GROUP BY: Line 227

Before (Class Analytics):

Instructor Work Load

Load

Instructor	Number of Classes Taught
------------	--------------------------

After:

Instructor Work Load

Load

Instructor	Number of Classes Taught
E002	1
E001	3
E003	2
E004	1
E005	1

Aggregation with HAVING: Line 240

Before (Class Analytics):

Low Enrollment

Find days with low enrollment (ie. days with less then half the average enrollment across all days.)

After: Load

Date	Enrolled
------	----------

After:

Low Enrollment

Find days with low enrollment (ie. days with less then half the average enrollment across all days.)

After: Load

Date	Enrolled
2024-12-23	1

Nested Aggregation with GROUP BY: Line 214

Before (Class Analytics):

Class Analytics

Average Class Size by Level

Load

Level	Average Class Size
-------	--------------------

After:

Class Analytics

Average Class Size by Level

Load

Level	Average Class Size
Intermediate	2.33333333333333
Advanced	1.5
Beginner	3.66666666666667

Division: Line: 299-313

Before (Lifts and Runs):

Lifts and Runs

Find Runs for Lift

Select a run difficulty to find lifts that can access it:

Find Lifts

After:

Lifts and Runs

Find Runs for Lift

Select a run difficulty to find lifts that can access it:

Find Lifts

Lift Number	Chair Capacity	Status
L001	4	Operational
L006	4	Under Maintenance